

Chapter 7: Embryogenesis and Fetal Development — Revision Notes

Williams Obstetrics 26e, pp. 330–382 of the PDF. High-yield numbers are in **bold**. The two tables are transcribed from the PDF images (they are missing from the extracted chapter text). 8 of the 11 figures are included from the PDF (embryo drawings/photos 7-3 and 7-5 and the 12-week CRL image 7-7 are omitted).

Big picture: The fetus is a patient. Know how gestational age is defined and dated, the embryonic (organogenesis) vs fetal (growth) periods, the fetal circulation and its changes at birth, organ-by-organ maturation milestones, fetal fuel supply and how the placenta transfers substances.

1. Gestational Age

Term	Starts from	Term length
Gestational (menstrual) age	First day of the LMP (~2 wk before ovulation, ~3 wk before implantation)	280 days = 40 wk (9 $\frac{1}{3}$ calendar months)
Ovulation age	Ovulation (used by embryologists)	266 days = 38 wk
Postconceptional age	Fertilization ($\approx$ ovulation age)	—

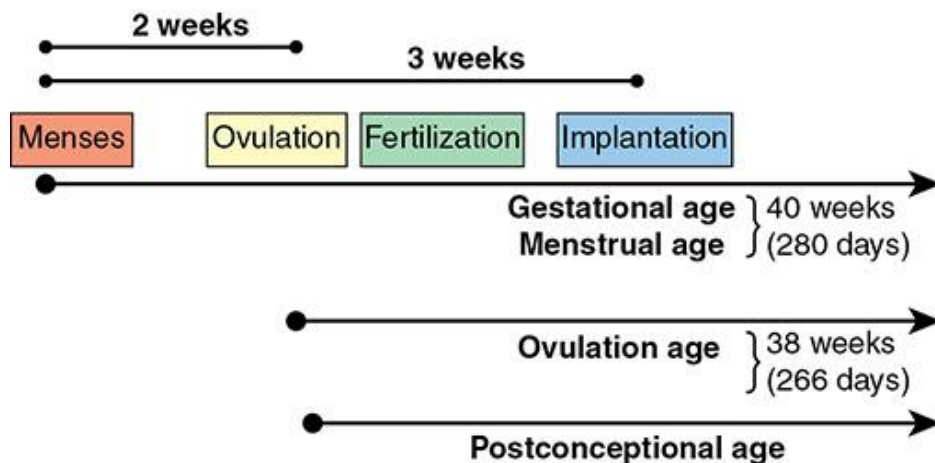


Figure 7-1. Gestational (menstrual) age counts from the LMP: 40 weeks (280 days); ovulation/postconceptional age is 2 weeks less (38 weeks, 266 days).

ACOG / AIUM / SMFM (2019) dating recommendations

1. **First-trimester sonography is the most accurate** way to establish or confirm gestational age.
2. **IVF:** use the **embryo age and transfer date**.
3. Compare LMP and first-trimester sonographic dates; record and discuss the **EDC**.
4. Record the **best obstetrical estimate** of gestational age at delivery on the birth certificate.
 - **CRL is accurate to ± 5 –7 days.** Change the EDC if the difference is **>5 days before 9 wk** or **>7 days later in the 1st trimester**.

Naegele rule

- **EDC = LMP + 7 days – 3 months.** E.g., LMP October 5 → **July 12**.
- **Three trimesters of ~14 wk each.** Wheels, EMR calculators and apps (e.g., the ACOG app, which uses sonographic criteria plus LMP or transfer date) also give the EDC.

2. Embryonic Development

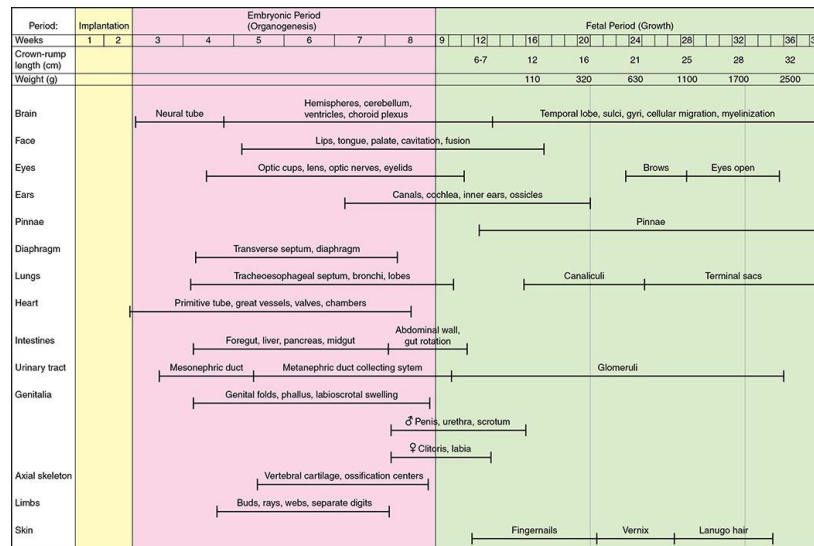


Figure 7-2. Organ-system timetable by menstrual weeks: implantation (wk 1–2), embryonic period of organogenesis (wk 3–8), then the fetal period of growth, with CRL and weight.

Fig 7-2's CRL and weight rows differ slightly from the text (e.g., 16 wk: 110 g vs 150 g; 24 wk: 630 g vs ~700 g; 32 wk: 1700 g vs ~1800 g; 36 wk: 2500 g vs ~2800 g; 12 wk CRL 6–7 cm vs 5–6 cm). The text values are used below.

Zygote and blastocyst

- First **2 weeks** after ovulation/fertilization = **preembryo** → blastocyst. **Implantation day 6–7**.
- 58-cell blastocyst** = **5 inner cell mass (embryo)** + **53 trophoblast**.

Embryonic period

- Conceptus is an **embryo from the start of the 3rd week after fertilization**; primitive villi form around the expected day of menses; most **hCG tests are positive** by now.
- Organogenesis lasts 6 weeks**, running through the **8th week**.

Timing	Milestone
Week 3 (postfertilization)	Body stalk differentiated; villous cores with angioblastic mesoderm; true intervillous space with maternal blood; fetal vessels appear in villi
Week 4	Cardiovascular system formed → true circulation (embryo ↔ villi); heart partitioning begins; neural plate folds into the neural tube ; limb buds
End of menstrual wk 5	Chorionic sac ~ 1 cm ; embryo 3 mm (measurable by sonography); arm and leg buds; amnion ensheathes the body stalk (→ cord)
End of wk 6	Embryo ~ 9 mm ; neural tube closed ; cardiac motion almost always visible
End of wk 8	CRL ~ 22 mm ; fingers and toes; elbows bend; upper lip complete; external ear elevations

- Neural tube closure: cranial end by 38 days and caudal end by 40 days from the LMP.**

The text's timing wording is inconsistent: it dates the embryo stage from the 3rd week after fertilization, but also says the period "begins the third week from the LMP". The fetal period starts at 7 wk after fertilization (9 wk LMP).

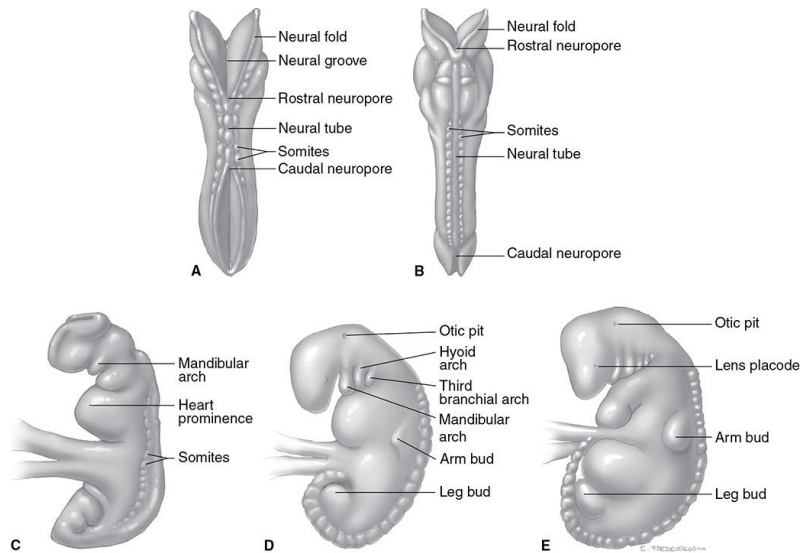


Figure 7-4. Embryos at 22–28 days: neural tube with rostral and caudal neuropores, somites, branchial arches, heart prominence, and arm and leg buds.

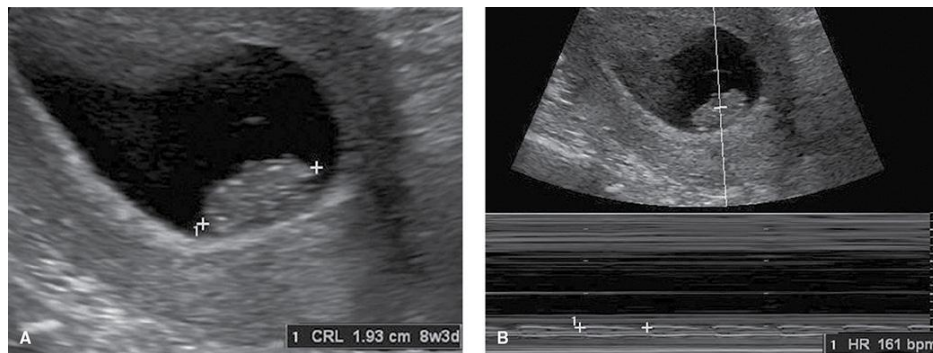


Figure 7-6. At 8 wk 3 d the CRL is 1.93 cm; M-mode already shows cardiac activity (161 bpm).

3. Fetal Period Epochs

- **Embryonic** → **fetal period at 7 wk after fertilization = 9 wk after LMP**. Fetus ~**24 mm**; most organs formed; now growth and maturation.

GA (wk)	CRL	Weight	Key features
12	5–6 cm	—	Uterus just palpable above the symphysis . Ossification centers in most bones; fingers and toes differentiated; skin, nails, hair rudiments; external genitalia beginning to show sex; spontaneous movements
16	12 cm	~150 g	Growth slows. CRL not measured after 13 wk (~8.4 cm) → BPD, HC, AC, FL instead. Eye movements 16–18 wk . Female: uterus formed and vaginal canalization begins by 18 wk ; male: testes start descending by 20 wk
20	—	>300 g	Midpoint . Linear weight gain. Moves ~every minute; active 10–30% of the day. Brown fat; lanugo ; scalp hair. Cochlear function 22–25 wk (matures 6 months after birth)
24	—	~700 g	Wrinkled skin, fat deposition begins; brows and lashes. Type II pneumocytes begin surfactant secretion . Canalicular period nearly complete but terminal sacs not yet formed . Survival barely >50% . Eyes open by 26 wk ; nociceptors and neural pain system developed
28	25 cm	~1100 g	Thin red skin with vernix caseosa ; pupillary membrane just gone; eye blinking peaks. Bone marrow = main site of hemopoiesis. 90% survival without impairment
32	28 cm	~1800 g	Skin still red and wrinkled
36	32 cm	~2800 g	Subcutaneous fat → rounder body, fuller face. Survival ~100%
40 (term)	36 cm	~3500 g	Fully developed

4. Central Nervous System

Brain

- **Neural tube closes: cranial end by 38 d, caudal end by 40 d from the LMP** → **folic acid must be started before this** to prevent NTDs.
- Walls → brain and spinal cord; lumen → ventricles and central canal.
- **Week 6**: three primary vesicles. **Week 7**: five secondary vesicles (Hox-controlled):

Vesicle	Becomes
Telencephalon	Cerebral hemispheres
Diencephalon	Thalami
Mesencephalon	Midbrain
Metencephalon	Pons, cerebellum
Myelencephalon	Medulla

- **Neuronal proliferation peaks at 3–4 months** (disorders here are devastating, e.g., **Zika**). **Migration peaks at 3–5 months** (ventricular/subventricular zones → final sites). Both **complete by 20–24 wk**.
- Then gyral formation and maturation; brain appearance tracks gestational age (MRI studies).
- **Myelination** of cerebrospinal nerve ventral roots and brainstem starts **~6 months**; most is **after birth**. Little myelin + incomplete skull ossification let the brain be seen on sonography.

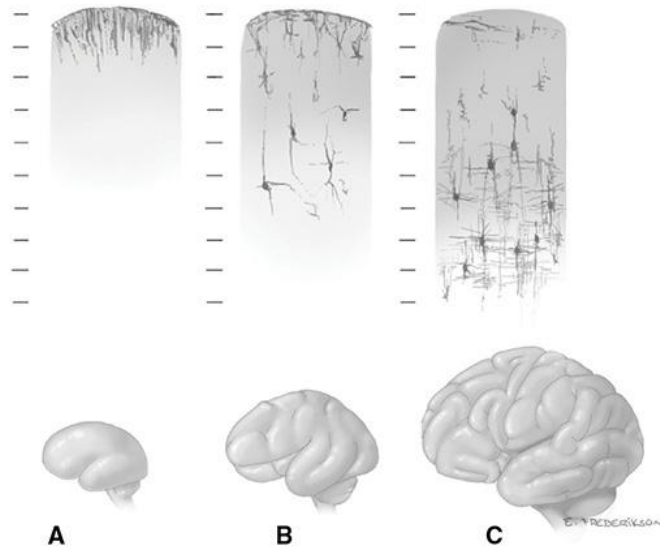


Figure 7-8. Cortical organization and gyral development at 20, 35 and 40 weeks; proliferation and migration are complete by 20–24 weeks.

Spinal cord

- Upper $\frac{2}{3}$ of the neural tube → brain; **lower $\frac{1}{3}$ → spinal cord.**
- **Entire sacrum ossified on sonography by ~21 wk.**
- **Conus level: S1 at 24 wk → L3 at birth → L1 in adults.**
- Cord myelination starts midgestation, continues through year 1. **Synaptic function by week 8** (neck and trunk flexion).

5. Cardiovascular System

- More than 100 genes/factors (e.g., **HIF, HOX**). **Straight heart tube by day 23. Weeks 4–7:** growth and remodeling → partially septated 4-chambered heart with primitive valves; aortic arch by vasculogenesis. Coronary angiogenesis late in fetal life.

Fetal circulation

- Placenta oxygenates the blood → most RV output **bypasses the lungs. Ventricles work in parallel**, not in series.
- **Umbilical vein** (single) → **ductus venosus** (major branch; straight to IVC with well-oxygenated blood) + **portal sinus** (to hepatic veins, mainly left liver; O_2 extracted).
- IVC blood = mix of ductus venosus blood + deoxygenated lower-body blood → lower O_2 content than blood leaving the placenta.

Streaming in the right atrium

Stream	Route	Destination
Well-oxygenated (dorsomedial IVC)	Crista dividens → foramen ovale → LA → LV	Heart and brain
Less oxygenated (lateral IVC wall), SVC and coronary sinus	Tricuspid → RV	Pulmonary trunk → ductus arteriosus → descending aorta

- **RV = $\frac{2}{3}$ of total cardiac output.** RV blood is **15–20% less saturated** than LV blood.
- **~90% of RV output passes through the ductus arteriosus;** only **~8% goes to the lungs** (high pulmonary resistance).
- Of ductus blood, **$\frac{1}{3}$ goes to the body;** the rest returns to the placenta via the **two hypogastric arteries** → umbilical arteries.

- SVC flow velocity rises from 20 wk to term.

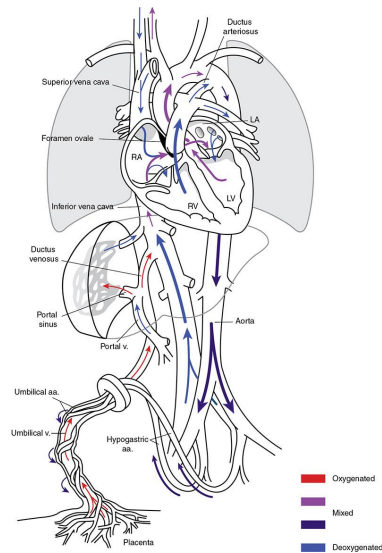


Figure 7-9. Fetal circulation: umbilical vein → ductus venosus → IVC → foramen ovale to the left heart; SVC blood → RV → ductus arteriosus; hypogastric (umbilical) arteries return blood to the placenta.

Circulatory changes at birth

Fetal structure	Change	Adult remnant
Ductus arteriosus	Functional closure; lungs expand → RV blood goes to the lungs	—
Foramen ovale	Closes	—
Distal hypogastric (umbilical) arteries	Atrophy/obliterate within 3–4 days	Umbilical ligaments
Umbilical vein (intraabdominal)	Obliterates	Ligamentum teres
Ductus venosus	Constricts 10–96 h ; anatomically closed by 2–3 wk	Ligamentum venosum

- Ventricles switch **almost instantly** from parallel to **series** function.

Fetoplacental blood volume

- Term newborn (immediate clamping) **~78 mL/kg** + placental fetal blood **~45 mL/kg** → **fetoplacental volume ~125 mL/kg** (used to size **fetomaternal hemorrhage**).

6. Hemopoiesis

- **Sites in order: yolk sac (and endothelium) → liver → spleen and bone marrow.** Marrow is the main site by **28 wk**.
- **Erythrocytes:** first ones are **nucleated and macrocytic**.
 - **MCV ≥ 180 fL** in the embryo → **105–115 fL at term** (adult 80–95 fL).
 - **Aneuploid fetuses keep a high MCV (~130 fL).**
 - Life span lengthens to **~90 days at term**. Reticulocytes fall to **4–5%** at term. More deformable (offsets size and viscosity); different enzyme activities.
- **SMFM: fetal anemia = hematocrit $< 30\%$.**

Table 7-1. Fetal Hemoglobin Concentrations Across Pregnancy (g/dL)

Weeks' Gestation	1.16 MoM (95th percentile)	1.00 MoM (median)	0.84 MoM (5th percentile)
18	12.3	10.6	8.9
20	12.9	11.1	9.3
22	13.4	11.6	9.7
24	13.9	12.0	10.1
26	14.3	12.3	10.3
28	14.6	12.6	10.6
30	14.8	12.8	10.8
32	15.2	13.1	10.9
34	15.4	13.3	11.2
36	15.6	13.5	11.3
38	15.8	13.6	11.4
40	16.0	13.8	11.6

MoM = multiples of the median.

Erythropoietin and platelets

- Erythropoiesis is driven by **fetal erythropoietin** (maternal EPO does **not** cross the placenta). Stimulated by testosterone, estrogen, prostaglandins, thyroid hormone, lipoproteins and above all **fetal hypoxia**.
- EPO rises with maturity; **fetal liver is the source until the kidney takes over near term**. Amnionic fluid EPO correlates with umbilical venous EPO. **May be undetectable for up to 3 months after birth**.
- Platelets reach stable levels by midpregnancy.**

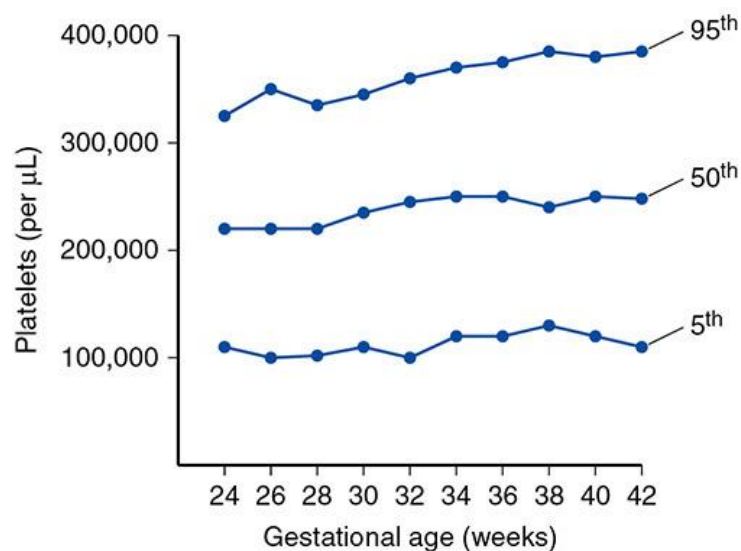


Figure 7-10. Platelet counts on day 1 of life by gestational age (24–42 wk): median ~220,000–250,000/µL, 5th percentile ~100,000–130,000/µL.

Fetal hemoglobin

- β-type genes on chromosome 11; α-type on chromosome 16.** Genes switch on and off (off by **methylation**).

Stage / site	Hemoglobins
I. Embryonic (yolk sac)	Gower 1 ($\zeta_2\varepsilon_2$), Gower 2 ($\alpha_2\varepsilon_2$), Portland ($\zeta_2\beta_2$)
II. Fetal (liver)	HbF ($\alpha_2\gamma_2$)
III. Adult (bone marrow)	HbA1 ($\alpha_2\beta_2$), HbA2 ($\alpha_2\delta_2$)

- **Adult α chain is made exclusively by 6 wk**; ζ is switched off $\rightarrow$ an **α -gene mutation has no substitute**.
- **γ and δ chains continue** $\rightarrow$ with a **β -gene mutation**, HbF ($\alpha_2\gamma_2$) or HbA2 ($\alpha_2\delta_2$) can substitute.
- **Persisting HbF (γ -gene hypomethylation)**: infants of **diabetic mothers**; **sickle cell anemia** ($\uparrow$ HbF = milder disease; basis of HbF-inducing drugs).
- **HbF binds O_2 more avidly than HbA** because **HbA binds 2,3-DPG more avidly**; maternal 2,3-DPG is $\uparrow$ in pregnancy and fetal RBCs have less 2,3-DPG.
- **At term $\sim 3/4$ of hemoglobin is HbF**; it falls to adult levels over **6–12 months**.

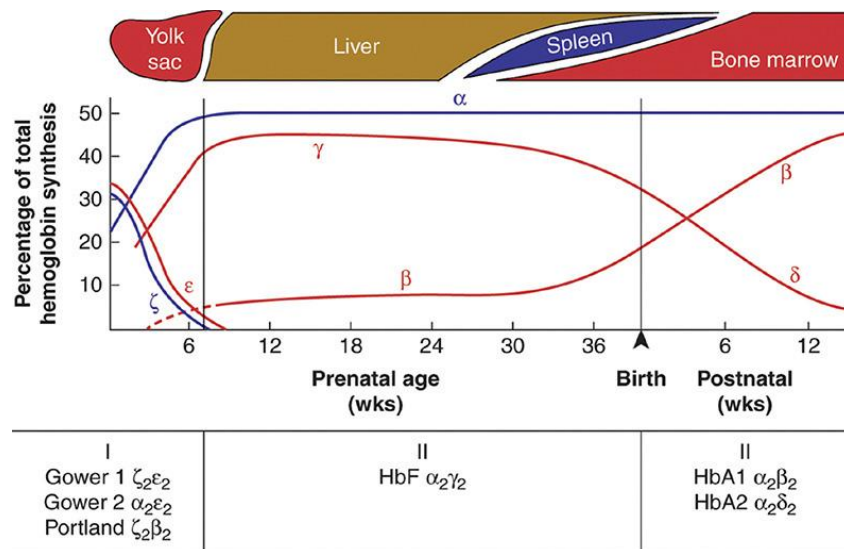


Figure 7-11. Globin switching: ζ and ε (yolk sac) give way to α and γ (liver, HbF); β rises after birth (HbA1). The postnatal panel is printed as stage "II"; it is stage III.

Coagulation

- Except **fibrinogen**, hemostatic proteins have no embryonic forms. Adult-type pro-, anti-coagulant and fibrinolytic proteins are made **from 12 wk**; they **don't cross the placenta**.

At birth $\approx$ 50% of adult	Close to adult levels
Factors II, VII, IX, X, XI; protein S, protein C, antithrombin, plasminogen	Factors V, VIII, XIII, fibrinogen

- **Vitamin K-dependent factors fall further in the first days** without prophylaxis (worse with **breastfeeding**) $\rightarrow$ **newborn hemorrhage**. Maternal vitamin K deficiency $\rightarrow$ fetal cerebral hemorrhage.
- **Fetal fibrinogen from 5 wk**: same amino acids but less compressible clot; lower plasma level but **functionally more active**.
- Functional **factor XIII significantly** $\downarrow$. Low plasminogen, $\uparrow$ fibrinolytic activity in cord plasma. **Cord platelet counts are normal adult range**.
- Despite this, **fetal bleeding is rare** (even after cordocentesis): amniotic fluid thromboplastins + a Wharton jelly factor aid clotting.
- Inherited thrombophilia (usually **homozygous**) $\rightarrow$ placental/fetal thrombosis; e.g., **homozygous protein C deficiency** $\rightarrow$ **purpura fulminans**.

Plasma proteins

- Made by the fetus; not correlated with maternal levels. **Albumin, LDH, AST, ALT, GGT rise**; **prealbumin falls** with GA.

- At birth, total protein and albumin $\approx$ maternal levels → **albumin binds unconjugated bilirubin** (prevents kernicterus).

7. Respiratory System

- Lung maturity predicts early neonatal outcome; immaturity → **respiratory distress syndrome**. Surfactant in amniotic fluid = biochemical maturity, but **structural maturation matters too**.
- Lung primordium from **foregut endoderm ~day 20**; **lung bud day 25**.

Stages of lung development

Stage	Timing	Events
Pseudoglandular	5–17 wk	Intrasegmental bronchial tree grows; microvasculature begins; looks like a gland
Canalicular	16–25 wk	Cartilage plates extend; terminal bronchioles → respiratory bronchioles → saccular ducts
Terminal sac	After 25 wk	Primitive alveoli (terminal sacs); ECM develops proximal → distal until term
Alveolar	Late fetal → childhood	Capillary network, lymphatics; type II pneumocytes make surfactant

- **Only ~15% of adult alveoli at birth**; alveoli added **up to age 8**.

Timing of insults

Insult	Effect
Embryonic-phase defects	Esophageal/tracheal atresia, TE fistula , pulmonary agenesis
Renal agenesis (no fluid from the start)	Major defects in all four stages
ROM + oligohydramnios before 20 wk	Near-normal bronchial branching and cartilage, but immature alveoli
ROM after 24 wk	Minimal long-term effect on lung structure

- Vitamin D is thought to aid lung development.

Surfactant

- Keeps terminal sacs open after the first breath. Made in **type II pneumocytes** (multivesicular bodies → **lamellar bodies**). Late in fetal life lamellar bodies are swept into amniotic fluid by **fetal breathing**; at the first breath surfactant uncoils and lines the alveolus.

Component	Share
Lipid (glycerophospholipids)	90% of dry weight
Protein	10%
Phosphatidylcholines (lecithins)	~80% of glycerophospholipids
Dipalmitoyl phosphatidylcholine (DPPC)	~½ of surfactant = principal active component
Phosphatidylglycerol (PG)	8–15% ; role unclear (newborns without PG usually do well)
Phosphatidylinositol (PI)	Other major constituent

- Phospholipid lowers surface tension; apoproteins help form the surface film.
- **Major apoprotein SP-A**: glycoprotein **28,000–35,000 Da**; amniotic fluid content rises with maturity; **gene expression by 29 wk**; **SP-A1 and SP-A2 on chromosome 10**.
- **Corticosteroids** (Liggins, lambs) accelerate lung maturation; not the only stimulus, but antenatal **betamethasone/dexamethasone** help at critical times (Ch 34).

Fetal breathing

- Chest wall movements on sonography from **11 wk**. **From the 4th month** strong enough to move fluid in and out. **Essential for lung growth**. Maternal exercise stimulates it.

8. Digestive System

Gut	Derivatives
Foregut	Pharynx, lower respiratory tract, esophagus, stomach, proximal duodenum, liver, pancreas, biliary tree
Midgut	Distal duodenum, jejunum, ileum, cecum, appendix, right colon
Hindgut	Left colon, rectum , upper anal canal

- Rotation, fixation and partitioning errors → malformations (e.g., intestinal atresias).
- **Swallowing begins 10–12 wk** (with peristalsis and active glucose transport). Preterm infants may swallow poorly.
- Fetus ingests **~800 mg soluble protein/day** late in pregnancy.
- **Term fetus swallows 200–760 mL/d** (like a neonate) → major regulator of fluid volume at term; **impaired swallowing → hydramnios**. Early on, swallowing has little effect on volume.
- Taste: **saccharin ↑ swallowing**, noxious chemicals ↓ it.
- **Intrinsic factor by 11 wk; pepsinogen by 16 wk**. Stomach emptying driven by volume; **a dilated stomach suggests obstruction**.
- **Anencephalic fetuses** (limited swallowing) often have **normal fluid volume** and GI tract → other regulators exist.

Meconium

- Contents: lung glycerophospholipids, desquamated cells, lanugo, scalp hair, vernix, swallowed debris. **Dark green-black from bile pigments (biliverdin)**.
- Passage: normal peristalsis in the mature fetus, **vagal stimulation**, or **hypoxia → fetal pituitary AVP → colonic contraction**.
- **Toxic to the lungs → meconium aspiration syndrome**.

Liver

- Hepatic diverticulum from foregut endoderm. **Liver = 10% of fetal weight at 9 wk**. Enzyme levels rise with GA; early hemopoiesis site.
- **Bilirubin**: limited, GA-dependent conjugation → preterm infants at risk of **unconjugated hyperbilirubinemia**. Short RBC life span → more bilirubin.
 - Most **unconjugated bilirubin → amniotic fluid (after 12 wk) and across the placenta** (transfer is **bidirectional**: maternal hemolysis → fetus and fluid).
 - **Conjugated bilirubin is not exchanged** significantly.
- **Cholesterol**: mostly fetal hepatic synthesis (for adrenal LDL demand); **20–50% maternal**. FGR fetuses have ↓ cholesterol from ↓ **fetal synthesis**.
- **Glycogen**: low in the 2nd trimester; rises near term to **2–3× adult liver levels**; falls sharply after birth.

Pancreas

- Dorsal and ventral buds from foregut endoderm.
- **Insulin granules by 9–10 wk; plasma insulin by 12 wk**. Fetal pancreas **secretes insulin in response to hyperglycemia**. Islets enlarged in fetuses of mothers with metabolic disease (pancreatic hyperechogenicity).
- **Glucagon at 8 wk**; fetal hypoglycemia doesn't raise it, but similar stimuli do by 12 h after birth.

- Trypsin, chymotrypsin, phospholipase A, lipase at **14 wk**; amylase in fluid at **14 wk**; most enzymes by **16 wk**. Exocrine function limited (needs a secretagogue, e.g., vagal acetylcholine). **No CCK** (needs protein ingestion).
- Sonographically visible in 60% at 19–36 wk.

9. Urinary System

- **Pronephros** involutes by **2 wk**; **mesonephros** makes urine at **5 wk** and degenerates by **11–12 wk**; **metanephros** (ureteric bud + nephrogenic blastema) **9–12 wk**.
- **Glomerular filtration begins by week 9. Loop of Henle functional by 14 wk. New nephrons until 36 wk** (continue after birth in preterm infants).
- Kidney and ureter from **intermediate mesoderm**; bladder and urethra from the **urogenital sinus** (bladder partly from the **allantois**).
- Fetal urine is **hypotonic**, low in electrolytes; limited concentrating and pH control. High renal vascular resistance, low filtration fraction. Regulated by RAS, sympathetic system, prostaglandins, kallikrein, ANP.
- **GFR: <0.1 mL/min at 12 wk → 0.3 mL/min at 20 wk**; constant per fetal weight later. **Hemorrhage or hypoxia ↓ renal flow, GFR and urine.**

GA	Urine output
12 wk	Urine production starts
18 wk	7–14 mL/d
Term	650 mL/d

- **Maternal furosemide ↑ fetal urine; uteroplacental insufficiency and FGR ↓ it.**
- Obstruction (urethra, bladder, ureters, pelves) damages parenchyma (Ch 19).
- **Kidneys are not essential in utero**, but **chronic anuria → oligohydramnios + pulmonary hypoplasia.**

10. Endocrine Glands

- The fetal endocrine system works before the CNS is mature.

Pituitary

- **Anterior** from oral ectoderm (**Rathke pouch**); **posterior** from **neuroectoderm**.
- Anterior lobe: 5 cell types, 6 hormones (lactotropes PRL, somatotropes GH, corticotropes ACTH, thyrotropes TSH, gonadotropes LH + FSH).
 - **ACTH at 7 wk; GH and LH by 13 wk; all pituitary hormones by end of wk 17.**
 - Fetal pituitary secretes **β-endorphin**; cord β-endorphin and β-lipotropin rise with **PaCO₂**.
 - **Intermediate lobe** well developed in the fetus; disappears before term (absent in adults).
- **Neurohypophysis** developed by **10–12 wk**: oxytocin and AVP present; conserve water mainly via **lung and placenta**, not kidney. **Cord AVP » maternal.**

Thyroid

- From **pharyngeal endoderm**; migrates; thyroglossal duct obliterates to the **foramen cecum**.
- **Pituitary–thyroid axis functional by end of 1st trimester. Hormone synthesis by 10–12 wk**; TSH, T4 and TBG in fetal serum by **11 wk**.
- **Placenta concentrates iodine on the fetal side; from 12 wk the fetal thyroid concentrates iodine more avidly than the mother's → maternal radioiodine or large iodine doses are hazardous after 12 wk.**
- Free T4, free T3 and TBG rise steadily. **At 36 wk vs adult: TSH higher, total and free T3 lower, T4 similar → pituitary feedback sensitivity develops late.**

- Thyroid hormone is needed by almost all tissues, **especially the brain**.
- **Thyroid agenesis**: growth is normal because **small amounts of maternal T4 prevent antenatal cretinism**; stigmata appear **after birth** → **universal newborn TSH screening** (state law).
- **Placenta deiodinates maternal T4/T3** → **reverse T3** (limits transfer).
- **Antibodies that cross: LATS, LATS-P, TSI** → **congenital hyperthyroidism**: large **goiter**, tachycardia, hepatosplenomegaly, hematological abnormalities, **craniosynostosis**, FGR; later perceptual-motor problems, hyperactivity, reduced growth.
- **After birth**: cooling → **TSH surge** → **T4 peaks at 24–36 h**; T3 rises almost simultaneously.

Adrenal

- **Medulla from neural crest ectoderm; cortex (fetal and adult) from intermediate mesoderm**.
- Growth driven by proliferation, angiogenesis etc.; **Kiss1R** ± CRH stimulate it.
- **10–20× larger than adult glands relative to body size**. Bulk = **fetal zone**, which **involutates rapidly after birth**; scant/absent with congenital absence of the pituitary. (Function: Ch 5.)

11. Immunological System

- Immune competence reported by **13 wk**. Cord blood components at term ≈ **½ adult values**.
- Without infection, fetal immunoglobulin is almost all **maternal IgG** → newborn antibodies reflect **maternal** experience.

Ig Key facts

IgG Transfer via placental **Fc receptors**, begins ~**16 wk**, bulk in the last 4 wk → **preterm infants poorly protected**. Newborn reaches adult IgG at **3 yr**. Harmful example: **Rh alloimmunization** (hemolytic disease)

IgM **Not transferred from the mother** → any fetal IgM is fetal. ↑ **cord IgM = congenital infection** (rubella, CMV, toxoplasmosis). Adult levels by **9 months**

IgA From **colostrum**: mucosal protection against enteric infection. Little fetal secretory IgA in amniotic fluid

- **B cells in fetal liver by 9 wk**, blood and spleen by **12 wk**. **T cells leave the thymus ~14 wk**.
- Newborns respond poorly to immunization, especially **bacterial capsular polysaccharides**. Newborn monocytes can present antigen.

12. Musculoskeletal System

- Mostly **mesodermal**. **MYOD** activates muscle-specific genes. **Limb buds by week 4**. Skeletal muscle from somite myogenic precursors.
- Skeleton from condensed mesenchyme → **hyaline cartilage models** → **endochondral ossification**; ossification centers by the end of the embryonic period. Osteoclasts from erythro-myeloid progenitors.

13. Energy and Nutrition

- Little yolk → embryo depends on maternal nutrients in the **first 2 months**; right after implantation the blastocyst feeds on endometrial interstitial fluid.
- **Maternal depots: liver, muscle, adipose**. Insulin drives storage (glycogen, protein, fat). **Maternal fat storage peaks in the 2nd trimester**, then falls as fetal demand rises in the 3rd.
- Placenta acts as a **nutrient sensor**. In fasting, glycogen is insufficient → **lipolysis** releases free fatty acids.

Glucose

- **Main fetal fuel.** From midpregnancy **fetal glucose is independent of, and may exceed, maternal levels.**
- **hPL** (abundant in mother, not fetus) is an **insulin antagonist**: blocks maternal glucose uptake, promotes FFA use → spares glucose for the fetus; **diabetogenic.**
- **Transfer: facilitated diffusion** (carrier-mediated, stereospecific, nonconcentrating). **14 GLUTs** (SLC2A family); **GLUT-1, -3, -4** in syncytiotrophoblast microvilli. GLUT expression regulated by **DNA methylation** (↑ with gestation).
- **Placental and fetal glucose use are inversely related.**
- **Lactate** crosses by facilitated diffusion (probably as lactic acid with H⁺).
- **Macrosomia: fetal hyperinsulinemia** is a key driver; IGF, leptin, adipokines regulate the placenta. **Maternal obesity begets macrosomia** (and may affect fetal cardiomyocytes). NAFLD is linked to macrosomia.

Leptin

- From mother, fetus and placenta (syncytiotrophoblast, fetal endothelium). Of placental leptin **5% → fetus, 95% → mother.**
- Amniotic fluid leptin peaks midpregnancy; **fetal serum leptin rises from ~34 wk and correlates with fetal weight**; ↓ in **FGR**. Abnormal levels with growth disorders, **GDM, preeclampsia** (not maternal obesity). Roles in heart, brain, kidney, pancreas maturation; later metabolic syndrome.

Fatty acids and triglycerides

- Term newborn fat **~15% of body weight.**
- **Triglycerides do not cross; glycerol does.** Abnormal maternal TG (low or high) → major anomalies.
- **Preferential transfer of long-chain polyunsaturated fatty acids.**
- **Lipoprotein lipase on the maternal side only** → hydrolysis in the intervillous space, neutral lipids preserved in fetal blood; fetal liver re-forms TG.
- **LDL (~250,000 Da)** taken up by **receptor-mediated endocytosis** in syncytial coated pits → lysosomal hydrolysis yields (1) **cholesterol for progesterone**, (2) **amino acids**, (3) **essential fatty acids (linoleic acid)**. High maternal cholesterol linked to fetal aortic atherogenesis.

Amino acids

- **Concentrated in syncytiotrophoblast**, then diffuse to the fetus → **cord plasma amino acids > maternal.**
- Transport modulated by GA, heat, hypoxia, under-/overnutrition, glucocorticoids, GH, leptin; **mTORC1** regulates transporters; ↑ transfer in **GDM with overgrowth.**

Proteins

- Large proteins mostly excluded, **except IgG** (endocytosis via trophoblast Fc receptors; depends on maternal IgG level, GA, placental integrity, subclass, antigenic potential). **IgA and IgM are excluded.**

Ions and trace metals

Substance	Transfer / fetal vs maternal level
Calcium, phosphorus	Active transport mother → fetus (skeletal mineralization); placental Ca-binding protein. PTH absent in fetal plasma; PTH-rP = the fetal parathormone
Iron	Unidirectional ; fetal plasma iron » maternal; fetal Hb normal even with severe maternal iron deficiency
Iodine	Active, carrier-mediated, energy-requiring ; placenta concentrates it
Zinc	Fetal > maternal
Copper	Fetal < maternal (sequestered by metallothionein)

Heavy metals

- **Metallothionein-1** in syncytiotrophoblast binds zinc, copper, lead, cadmium.
- **Lead reaches ~90% of maternal levels; cadmium transfer is limited** (main source: **cigarette smoke**). Cadmium may induce amnion metallothionein → copper sequestration → ↓ **amnion tensile strength**.

Vitamins

Vitamin	Fetal vs maternal
A (retinol)	Fetal > maternal ; bound to retinol-binding protein and prealbumin
C	Fetal 2–4× maternal (energy-dependent carrier)
D (incl. 1,25-(OH) ₂ D)	Maternal > fetal ; 1β-hydroxylation of 25-OH-D ₃ occurs in placenta and decidua

14. Placental Role in Embryofetal Development

- Bidirectional exchange: O₂ and nutrients to the fetus; CO₂ and wastes to the mother. Fetal blood in villous capillaries never directly contacts maternal intervillous blood.
- **Breaks in villi** → fetal cells enter maternal blood → **D sensitization**; **microchimerism** (1–6 cells/mL at midpregnancy; some persist for life) → may provoke **Hashimoto thyroiditis**.
- **Cell-free DNA** from syncytiotrophoblast turnover: **after 10 wk, 10–15% of maternal plasma cfDNA is trophoblastic** → basis of **cfDNA aneuploidy screening**.

Intervillous space

- Maternal spiral artery blood bathes the trophoblast; villi + intervillous space act as the **fetal lung, gut and kidney**.
- **Residual intervillous volume at term ~140 mL. Uteroplacental flow near term 700–900 mL/min**, mostly to the intervillous space.

Placental transfer

- Barrier in terminal villi: **syncytiotrophoblast** → **attenuated cytotrophoblast** → **villous stroma** → **fetal capillary wall**. An active, selective interface: maternal-facing **microvilli**, fetal-facing **basal membrane**, and villous capillaries.

Table 7-2. Variables of Maternal-Fetal Substance Transfer

Variable
Maternal plasma concentration
Maternal carrier-protein binding
Maternal blood flow rate through the intervillous space
Trophoblast surface area size available for exchange
Physical trophoblast properties to permit simple diffusion
Trophoblast biochemical machinery for active transport
Substance metabolism by the placenta during transfer
Fetal intervillous capillary surface area size for exchange
Fetal blood concentration of the substance
Fetal carrier-protein binding
Villous capillary blood flow rate
DNA methylation of transporter genes

Mechanisms of transfer

Mechanism	Examples
Simple diffusion (most substances <500 Da)	O₂, CO₂, water, most electrolytes, anesthetic gases
Facilitated by syncytiotrophoblast	Low-concentration essential compounds (e.g., glucose)
Very slow crossing	Insulin, steroid hormones, thyroid hormones
Receptor-mediated (large molecules)	IgG (160,000 Da) via trophoblast receptor
Unequal secretion of placental hormones	hCG, hPL much lower in fetal than maternal plasma

Oxygen and carbon dioxide

- **O₂ transfer is blood-flow limited.** Delivery **~8 mL O₂/min/kg** fetal weight.
- **Intervillous blood: O₂ saturation 65–75%, PaO₂ 30–35 mm Hg.** Umbilical vein saturation similar, PO₂ slightly lower. **HbF's higher O₂ affinity** helps.
- **CO₂ diffuses faster than O₂.** Umbilical artery **PaCO₂ ~50 mm Hg, ~5 mm Hg above** intervillous blood. Fetal blood has **less CO₂ affinity**, and **maternal hyperventilation ↓ PaCO₂** → both favor fetus → mother CO₂ transfer.

Memory hook: fetal O₂ loading = "**low 2,3-DPG + HbF + maternal hyperventilation**"; CO₂ unloading = "**fetus 50, intervillous 45**".

Quick-Recall: Developmental Timeline (menstrual weeks unless stated)

When	Event
Day 20 / 25	Lung primordium / lung bud
Day 23	Straight heart tube
Wk 4 (postfertilization)	Cardiovascular system and neural tube form; limb buds
38 / 40 days	Cranial / caudal neuropore closure
End wk 5	Embryo 3 mm, sac 1 cm
End wk 6	Embryo 9 mm; cardiac motion
6 wk	Adult α -globin only
7 wk	ACTH in pituitary; five brain vesicles
8 wk	CRL 22 mm; glucagon; synaptic reflexes
9 wk	Fetal period begins (24 mm); liver 10% of body weight; glomerular filtration; B cells in liver
10–12 wk	Swallowing; neurohypophysis; thyroid hormone synthesis; insulin granules
11 wk	Intrinsic factor; fetal TSH, T4, TBG; fetal breathing movements
12 wk	CRL 5–6 cm; urine; plasma insulin; coagulation proteins; fetal thyroid concentrates iodine
13 wk	Last CRL measurement (8.4 cm); GH and LH; immune competence
14 wk	Loop of Henle; T cells leave thymus; pancreatic enzymes
16 wk	150 g; eye movements (16–18); IgG transfer begins; pepsinogen
17 wk	All pituitary hormones
20 wk	>300 g; midpoint; neuronal migration complete (20–24)
21 wk	Sacrum ossified on sonography
24 wk	~700 g; surfactant secretion; spinal cord at S1; survival ~50%
26 wk	Eyes open
28 wk	1100 g; bone marrow hemopoiesis; 90% intact survival
29 wk	SP-A gene expression
34 wk	Fetal leptin rises
36 wk	2800 g; nephrogenesis complete
40 wk	36 cm, 3500 g; ~ $\frac{3}{4}$ HbF

Key Numbers to Memorize

- Term **280 d / 40 wk** (LMP) = **266 d / 38 wk** (ovulation); Naegele **+7 d –3 mo**
- CRL accuracy **± 5 –**7 d**; re-date if **>5 d before 9 wk** or **>7 d** later in 1st tri**
- Fetal period from **9 wk LMP (7 wk postfertilization)**, 24 mm
- Weights: 16 wk **150 g**, 20 wk **>300 g**, 24 wk **700 g**, 28 wk **1100 g**, 32 wk **1800 g**, 36 wk **2800 g**, 40 wk **3500 g**
- Neural tube closed by **38–40 days**; spinal cord **S1 (24 wk) → L3 (birth) → L1 (adult)**
- RV **$\frac{2}{3}$** of output; ductus arteriosus **~90%** of RV output; lungs **~8%**; RV blood **15–20%** less saturated
- Ductus venosus constricts **10–96 h**, closed **2–3 wk**; hypogastric arteries obliterate **3–4 d**
- Fetoplacental blood volume **~125 mL/kg** (78 newborn + 45 placenta)
- MCV **$\geq 180 \rightarrow 105$ –**115 fL** (aneuploid **~130**); RBC life span **~90 d**; reticulocytes **4–5%**; anemia = Hct **<30%**; median fetal Hb **10.6 g/dL at 18 wk → 13.8 at 40 wk****
- HbF **~ $\frac{3}{4}$** at term; adult α by **6 wk**; neonatal factors II, VII, IX, X, XI **~50%**
- Only **15%** of alveoli at birth; alveoli added to **age 8**; DPPC **$\frac{1}{2}$** of surfactant; PG **8–15%**; SP-A gene **29 wk**

- Swallowing **200–760 mL/d** at term; urine **7–14 mL/d at 18 wk** → **650 mL/d** at term; GFR **<0.1** → **0.3 mL/min** (12 → 20 wk)
- Fetal adrenal **10–20×** adult size relative to body; T4 surge peaks **24–36 h** after birth
- IgG from **16 wk**, mostly last **4 wk**; adult IgG **3 yr**, IgM **9 mo**
- Newborn fat **15%**; leptin **95%** to mother; lead **90%** of maternal; vitamin C fetal **2–4×**
- cfDNA **10–15%** fetal after 10 wk; microchimerism **1–6 cells/mL**
- Intervillous volume **140 mL**; uteroplacental flow **700–900 mL/min**; O₂ delivery **8 mL/min/kg**; intervillous SaO₂ **65–75%**, PO₂ **30–35 mm Hg**; UA PCO₂ **50 mm Hg**; diffusion **<500 Da**; IgG **160,000 Da**